

Homework 6

Please read the beginning of Section 1.4 in Durrett, through the end of Example 1.11. You probably worked out this example with your group in class on September 4th.

1. In the Durrett reading for Homework 5, you saw an example of two events that were not independent. In Example 1.7 from Section 1.3, Durrett's chance experiment is drawing two cards from a deck. Event A is "the first card is a spade." Event B is "the second card is a spade." Durrett writes that $P(A) = 1/4$ and $P(B) = 1/4$. Is that right?
 - a. Describe a suitable sample space Ω for this chance experiment. There are many reasonable choices for Ω . To make your thinking clear, you need to indicate what the individual outcomes in Ω are and how you will denote them. I recommend using a sample space where the outcomes are equally likely.

We can use the set of permutations of two cards taken from the set of 52 cards. An example of an outcome is (7H, QS), meaning that the first card is the seven of hearts and the second card is the queen of spades. There are $(52)(51)$ outcomes in this sample space, we treat them all as equally likely, which is reasonable if the deck is well shuffled.

- b. Use your sample space to calculate $P(A)$ and $P(B)$. Do you get $P(A) = 1/4$ and $P(B) = 1/4$?

We use the multiplication rule to count outcomes where the first card is a spade. We choose any spade for the first card (13 possibilities). Then we choose any remaining card for the second card (51 possibilities). By the multiplication rule, there are $(13)(51)$ such outcomes, and so

$$P(A) = \frac{(13)(51)}{(52)(51)} = 1/4.$$

We count outcomes where the second card is a spade in the same way. We choose any spade for the second card (13 possibilities). Then we choose any remaining card for the first card (51 possibilities). By the multiplication rule, there are $(13)(51)$ such outcomes, and so

$$P(B) = \frac{(13)(51)}{(52)(51)} = 1/4.$$

There are many other ways to reason out $P(A)$ and $P(B)$. Here is one that Durrett might prefer:

Let's calculate $P(B)$. There are 52 cards that could be our second pick. Every card is equally likely to be picked second. The events "second pick is x ", where x runs over the 52 cards in the deck, are disjoint. Therefore, for any card x , $P(\text{second pick is } x) = 1/52$. There are 13 spades in the deck, and so $P(B) = P(\text{second pick is one of the 13 spades}) = 1/52 + 1/52 + \dots + 1/52$ (with 13 terms $1/52$ added up), and so $P(B) = 13/52 = 1/4$.

2. In Homework 5, you solved Durrett's Exercise 28, about three independent events A, B, C with $P(A) = 1/4$, $P(B) = 1/3$, $P(C) = 1/2$. Let N be the random variable that counts how many of the events happen. Describe the distribution of N as follows:

a. What are the possible values of N ?

There are three events, and so the possible values for N are 0, 1, 2, and 3.

b. For each possible value i , of N , calculate $P(N = i)$, where " $N = i$ " is the event that exactly i of the events happen. Your work in Homework 5 gave the value $P(N = 1)$. Use the same methods to calculate probabilities in the other cases.

We use the algebra of events and independence as in Homework 5.

$P(N = 0)$:

The event $(N = 0)$ is $A^c \cap B^c \cap C^c$. Since A, B, C are independent, so are A^c, B^c , and C^c . Therefore, $P(N = 0) = (3/4)(2/3)(1/2) = 1/4$

$P(N = 1)$:

You calculated this value in Homework 5. See solutions for details of the calculation. $P(N = 1)$ is $11/24$.

$P(N = 2)$:

The event $(N = 2)$ is $(A \cap B \cap C^c) \cup (A \cap B^c \cap C) \cup (A^c \cap B \cap C)$. The events $A \cap B \cap C^c$ and $A \cap B^c \cap C$ are disjoint, since any common outcome would be in B and B^c . The events $A \cap B \cap C^c$ and $A^c \cap B \cap C$ are disjoint, since any common outcome would be in A and A^c . The events $A \cap B^c \cap C$ and $A^c \cap B \cap C$ are disjoint, since any common outcome would be in A and A^c . Therefore, by one of the rules of probability, we have

$$P((A \cap B \cap C^c) \cup (A \cap B^c \cap C) \cup (A^c \cap B \cap C)) = P(A \cap B \cap C^c) + P(A \cap B^c \cap C) + P(A^c \cap B \cap C).$$

Since A, B, C are independent, so are A, B, C^c ; A, B^c, C ; and A^c, B, C . We thus have

$$P(A \cap B \cap C^c) = (1/4)(1/3)(1/2) = 1/24$$

$$P(A \cap B^c \cap C) = (1/4)(2/3)(1/2) = 1/12$$

$$P(A^c \cap B \cap C) = (3/4)(1/3)(1/2) = 1/8$$

$$\text{and hence } P(N = 2) = 1/24 + 1/12 + 1/8 = 6/24 = 1/4$$

$P(N = 3)$:

The event $(N = 3)$ is $A \cap B \cap C$. Since A, B, C are independent, we have $P(N = 3) = (1/4)(1/3)(1/2) = 1/24$.

We can check this work by adding the probabilities for all values of N. We get $1/4 + 11/24 + 1/4 + 1/24 = 1$, as we would predict.

3. Please work Exercise 30 from Durrett. In this question, he is really asking you to calculate the distribution of the random variable N counting how many of the couples show up to dinner.

The procedure is the same as for question 2. I won't repeat all the reasoning, but I'll give the probabilities in a form that matches the method described in detail above.

$$P(N = 0) = (0.1)(0.11)(0.25) = 0.00275$$

$$P(N = 1) = (0.9)(0.11)(0.25) + (0.1)(0.89)(0.25) + (0.1)(0.11)(0.75) = 0.05525$$

$$P(N = 2) = (0.9)(0.89)(0.25) + (0.9)(0.11)(0.75) + (0.1)(0.89)(0.75) = 0.34125$$

$$P(N = 3) = (0.9)(0.89)(0.75) = 0.60075$$

As in the previous exercise, these probabilities add to 1.

4. Please work Exercise 31 from Durrett. The back of the book has a numerical answer: 0.4032. (That value looks slightly off to me.) Explain how to calculate the probability, including any implicit independence assumptions that Durrett wants you to make.

Let A_i be the event that the student gets all A's in semester i , where $i = 1, 2, 3, 4, 5, 6, 7, 8$. Since grades in individual classes are independent, for any i , $P(A_i) = (1/2)^4 = 1/16$.

Since grades in different classes are independent, the events A_i are independent. (Writing that out carefully would involve a lot of notation. I hope it seems plausible.) We want to calculate the probability that at least one of these events occurs, i.e. of the event $A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6 \cup A_7 \cup A_8$. We use our usual "at least" trick. According to the algebra of events (one of De Morgan's laws), $(A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6 \cup A_7 \cup A_8)^c$ is $A_1^c \cap A_2^c \cap \dots \cap A_8^c$, and so

$$P(A_1 \cup A_2 \cup \dots \cup A_8) = 1 - P(A_1^c \cap A_2^c \cap \dots \cap A_8^c).$$

Since the events A_i are independent, so are their complements. Therefore,

$$P(A_1^c \cap A_2^c \cap \dots \cap A_8^c) = (1 - 1/16)^8.$$

Calculating decimal approximations, I get $P(A_1 \cup A_2 \cup \dots \cup A_8) \approx 0.4032805262$, which rounds up to 0.4033, a bit more than what Durrett put in the back of the book.

5. Please work Exercise 33 from Durrett. The back of the book only has a value for the second probability (0.4914). Explain how to calculate both probabilities, including the independence assumptions that you need to make.

We use the “at least” trick as in the previous problem. We assume that all die rolls are independent (just as we assumed all class grades were independent).

The probability of not getting any 6's in four rolls is $(5/6)^4$. The probability of getting at least one six is thus $1 - (5/6)^4 \approx 0.5177$. The Chevalier de Méré made money on that bet in the long run, since his probability of winning was more than 50%.

The probability of getting a double 6 on a roll of two dice is $1/36$ (assuming the outcomes of the rolls are independent). The probability of not getting any double 6's in 24 tries is $(35/36)^{24}$. The probability of getting at least one double 6 in 24 tries is thus $1 - (35/36)^{24} \approx 0.4914$. Since that probability is less than 50%, he lost money in the long run on this bet.

6. Please work Exercise 35 from Durrett. Here is what the back of the book says for this exercise:

0:6/36, 1:10/36, 2:8/36, 3:6/36, 4:4/36, 5:2/36

In your solution, explain what Durrett's notation means. Show how to calculate the probabilities needed for the distribution of $|X - Y|$ using the familiar sample space for dice rolling from Durrett's Example 1.2 and used in many class and homework problems.

There are five possible values for $|X - Y|$, namely 0, 1, 2, 3, 4, 5. We write the outcomes for the event that each of these values occurs.

$|X - Y| = 0$. $\{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$. The probability is 6/36.

$|X - Y| = 1$. $\{(1,2), (2,1), (2,3), (3,2), (3,4), (4,3), (4,5), (5,4), (5,6), (6,5)\}$. The probability is 10/36.

$|X - Y| = 2$. $\{(1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4)\}$. The probability is 8/36.

$|X - Y| = 3$. $\{(1,4), (2,5), (3,6), (4,1), (5,2), (6,3)\}$. The probability is 6/36.

$|X - Y| = 4$. $\{(1,5), (2,6), (5,1), (6,2)\}$. The probability is 4/36.

$|X - Y| = 5$. $\{(1,6), (6,1)\}$. The probability is 2/36.